



PEDIATRIC SEIZURE

Effective: January 1, 2026

Replaces: January 1, 2025

1. Patient Care Goals

- 1.1. Prompt cessation of seizures in the prehospital setting
- 1.2. Identify Status Epilepticus characterized by continuous seizure lasting more than 5 minutes OR more than one seizure without return to baseline mental status
- 1.3. Minimizing adverse events in the treatment of seizures in the prehospital setting
- 1.4. Minimizing seizure recurrence during transport
- 1.5. Ensure no prolonged periods of hypoventilation during seizure activity

2. BLS Treatment

- 2.1. Routine Medical Care – Pediatric (700-S05)
 - 2.1.1. **Oxygen** – titrate as appropriate
- 2.2. Protect patient during active seizure
- 2.3. Spinal Motion Restriction as appropriate (700-M11)
- 2.4. Recovery position with head elevated, if no suspected spinal injury or mechanism
- 2.5. Consider cooling measures if febrile (loosen blankets and/or remove excessive clothing)
- 2.6. **Blood Glucose Level (BGL)**, if necessary treat per protocol (700-P03)

3. ALS Treatment (for active seizure and/or status epilepticus)

- 3.1. Active Seizure without Vascular Access:
 - 3.1.1. Do not delay treatment of active seizure to obtain glucose level or begin cooling measures
 - 3.1.2. Administer **Midazolam 0.2 mg/kg IM**, not to exceed a total single dose of 10 mg
 - 3.1.3. After IM Midazolam, establish **Vascular Access (IV)**, TKO
 - 3.1.4. If unable to establish IV access and patient has continued seizure after 5 minutes, may repeat **Midazolam 0.2 mg/kg IM** once, not to exceed a total single dose of 10 mg
- 3.2. Active Seizure with Vascular Access (IV):
 - 3.2.1. **Midazolam 0.1 mg/kg IV**, (slowly titrated to effect in increments of no more than 2.5 mg; not to exceed a total dose of 5 mg) may repeat once if seizure has not resolved after 5 minutes
- 3.3. Post Seizure:
 - 3.3.1. **Vascular Access (IV)**, TKO
 - 3.3.2. If patient has a recurrent seizure; **Midazolam 0.1 mg/kg IV**, may repeat once if seizure has not resolved after 5 minutes
- 3.4. **BASE CONTACT**: More than two doses of Midazolam are needed

4. Pertinent Assessment Findings

- 4.1. Patients with high fever/severe infection may present with violent uncontrollable shivering (rigors), teeth chattering or shaking. These patients will typically be conscious although potentially confused. Differentiate between active seizure and rigors through level of consciousness, incontinence, oral trauma, abnormal breathing, and past medical history.



- 4.2. For new onset of seizure or seizures refractory to medication administration, assess for other causes of seizure including trauma, stroke, electrolyte abnormality, toxic ingestion/withdrawal, pregnancy with eclampsia, fever/infection, and syncope.

5. Key Documentation Elements

- 5.1. Active seizure(s) witnessed by care providers during patient care
 - 5.1.1. Duration and type of seizure witnessed
- 5.2. Presence of eye deviation during/after seizure
- 5.3. Presence of periods of apnea, cyanosis, vomiting, bowel/bladder incontinence, or fever
- 5.4. Medications administered by non-EMS personnel
- 5.5. Neurologic status including GCS, pupil size, focal neurological deficits, signs of stroke
- 5.6. Blood glucose level